

Bitcoin analysis project

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1 Introduction

This project is devoted to the analysis of basic Bitcoin blockchain properties and was done in order to deepen my understanding of them. Many variables were extracted from the blockchain in order to examine their distributions and relations, for example:

- *height*
- *nTx*
- *weight*
- *size*
- *transaction fee*
- *time*
- *vsize*

2 Data

The blockchain data was extracted using Bitcoin Core RPC with `getblock()` at verbosity level 2. This method provides detailed block and transaction information.

Market price data was obtained from Binance API and aligned to the block origin times. The price per block graph follows below.

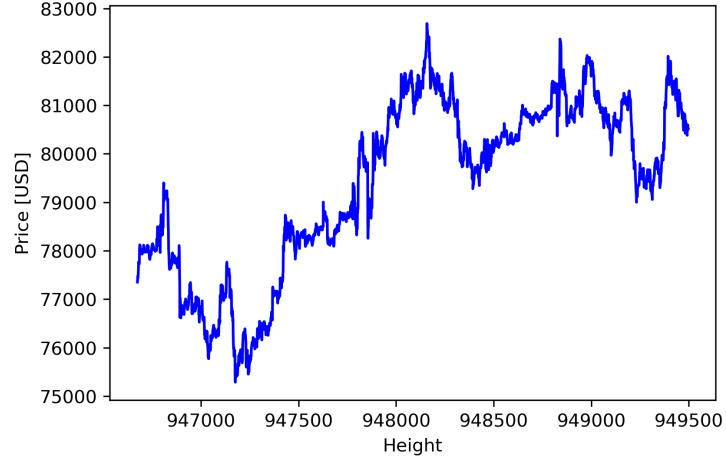


Figure 1: Price per each block

The number of blocks in the analysis is around 3000, since a pruned Bitcoin node was used for the analysis.

3 Basic quantities

At the beginning of the analysis, several histograms of basic quantities are presented.

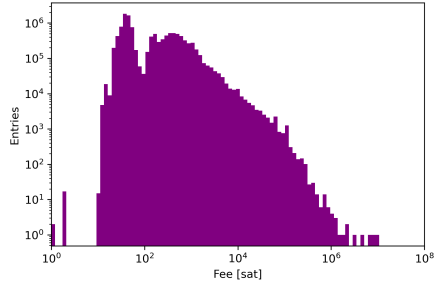


Figure 2: Histogram of fee distribution [sat]

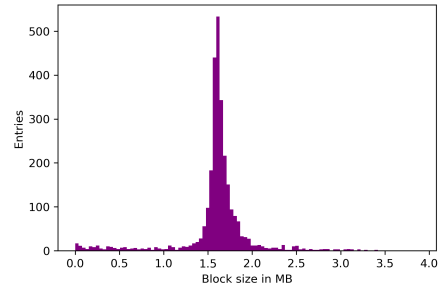


Figure 3: Histogram of block size in MB

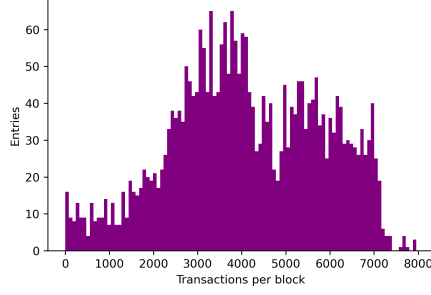


Figure 4: Histogram of transaction density

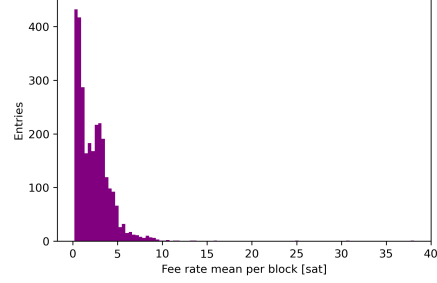


Figure 5: Histogram of fee rate mean

4 Block time distribution

Next, a histogram of block time distribution was created. If block production follows a stochastic Poisson process, block times are expected to follow an exponential distribution. The exponential distribution was fitted to the empirical data to test the hypothesis that block arrivals follow a Poisson process. The result of this is in 6. The function used for fit is

$$f(t) = \lambda e^{-\lambda t}, t \geq 0, \quad (1)$$

where λ is the rate at which the new blocks appear. The average time between blocks from this fit can be determined by $1/\lambda$. For the whole dataset, this average is 590.28s. This is consistent with the expected average block time of 600 seconds.

This average was also determined for the two epochs included in the dataset:

- Epoch 469: 606.17s
- Epoch 470: 583.50s

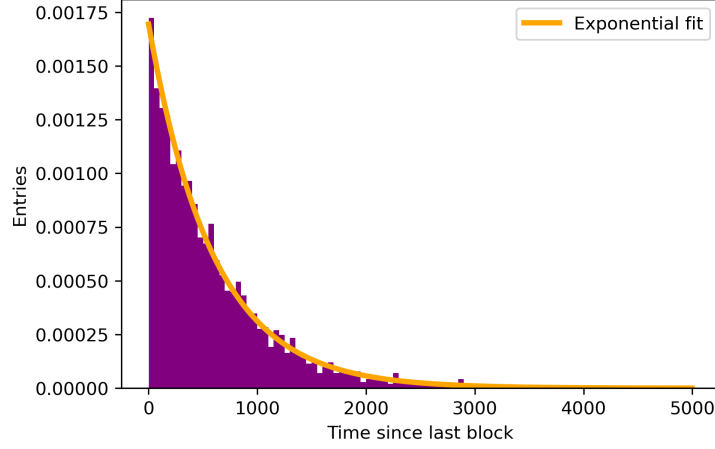


Figure 6: Poisson fit of the block time distribution.

As is visible from the plot, the Poisson function depicts the block time distribution well.

5 Fee rate distributions

The fee rate is calculated as

$$\text{fee_rate} = \frac{\text{fee}}{\text{vsize}}, \quad (2)$$

where the vsize is the virtual size introduced with the SegWit weight metric in 2017. Its purpose is to reduce the effective cost of witness data and thereby incentivise the adoption of SegWit transactions. The fee rates were analysed in several ways. First, the dependence of mean and median fee rates on block time is examined.

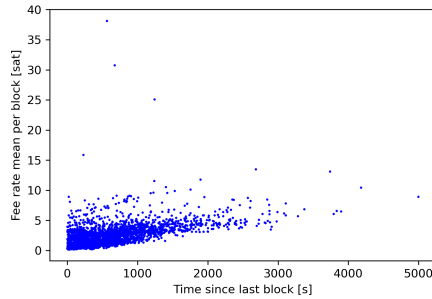


Figure 7: Fee rate mean per block as a function of block time

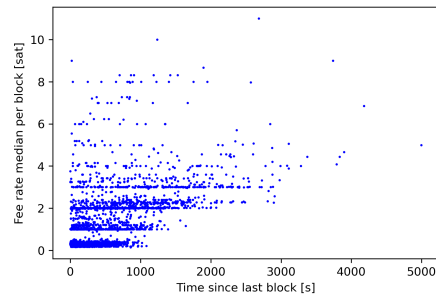


Figure 8: Fee rate median per block as a function of block time

These horizontal bands are observed only in the median fee rate plot and arise due to the discrete nature of fee rates, which results from integer-valued fees divided by transaction virtual sizes. Combined with a limited number of transactions per block, this leads to repeated identical median values across multiple blocks.

Then the focus was shifted to fee rate mean and fee rate median division. This plot suggests that a small number of high fee-rate transactions can significantly increase the mean relative to the median, indicating a heavy-tailed distribution within blocks.

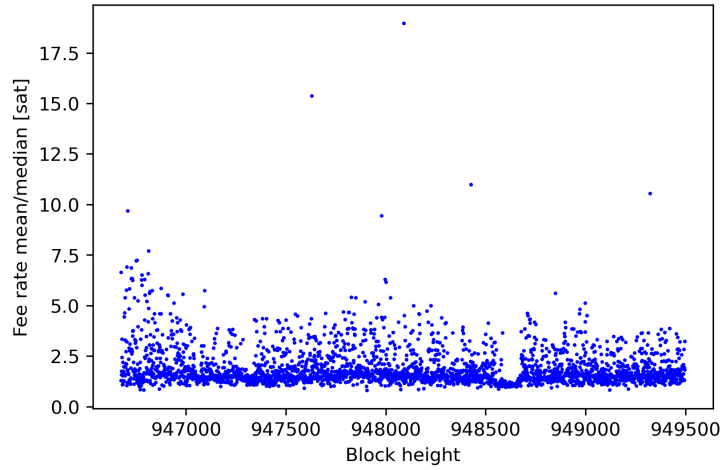


Figure 9: Ratio of mean to median fee rate per block

6 Block fill ratio

Block fill ratio was calculated as:

$$\text{fill ratio} = \frac{\text{weight}}{4 \cdot 10^6}, \quad (3)$$

thus the full block corresponds to a block with `weight` equal to $4 \cdot 10^6$ b. It is defined as

$$\text{weight} = 4 \cdot (\text{non witness bytes}) + 1 \cdot (\text{witness bytes}). \quad (4)$$

The weight metric was introduced together with the SegWit upgrade in 2017, as already mentioned in the preceding section.

From the plot that is presented below, it is evident, that most of the blocks have fill ratio close to 1, which indicates a high utilization of block space.

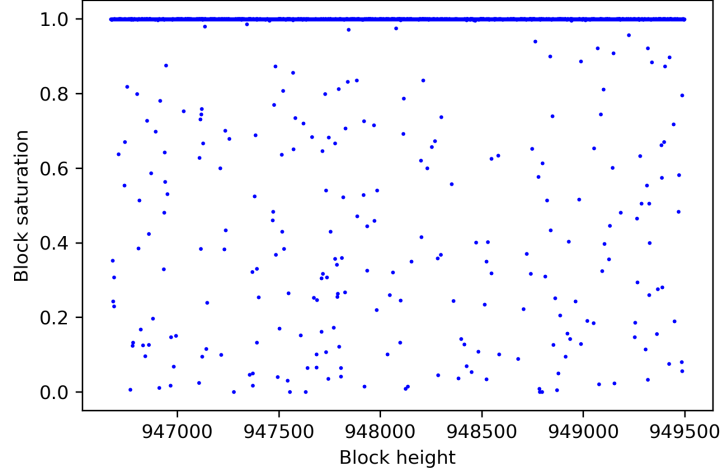


Figure 10: Block fill ratio plot.

7 Block time rolling average for two epochs in the dataset

Next distribution examined was the rolling average of the block time over 100 blocks. This was done for all the blocks together and then separated by epochs. The distributions differ a bit at the position of the target change since the minimum block rolling average count was set to 20 and was restarted at the position of new epoch in the second case.

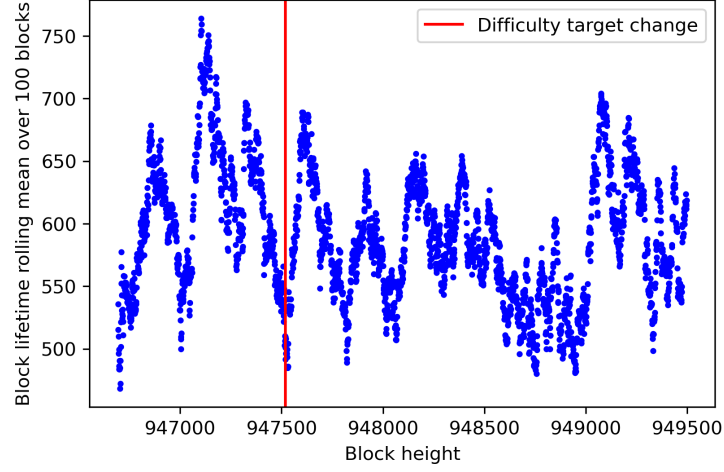


Figure 11: Block time rolling average over 100 blocks separated for two epochs in the dataset.

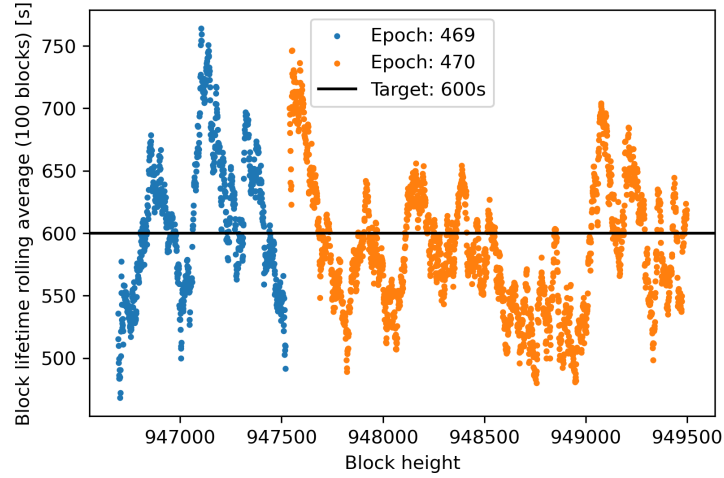


Figure 12: Block time rolling average over 100 blocks separated for two epochs in the dataset.

8 Correlation plot

At the end of the analysis, a simple plot depicting the correlation of some of the used quantities was created. Namely, these quantities are: Block time of the previous block, number of transactions, virtual size, block filling ratio (noted as saturation), price and fee rate median.

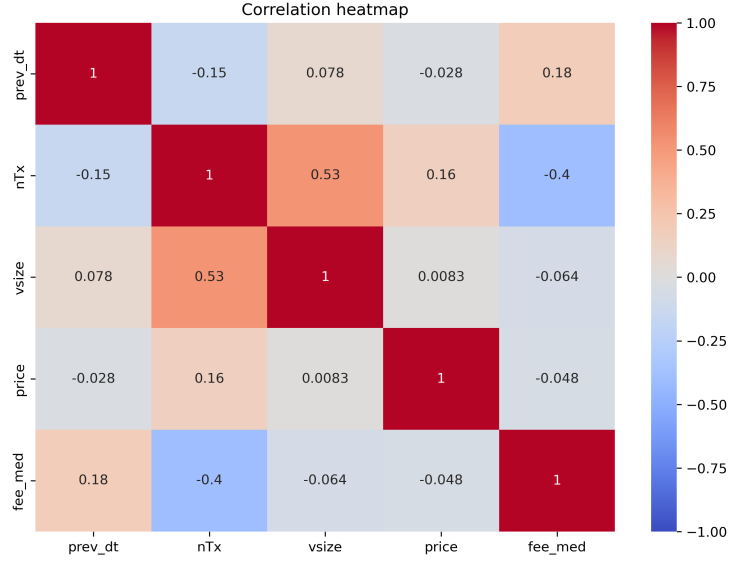


Figure 13: Correlation plot of several chosen quantities

As expected, the correlation between nTx and vsize is 0.53. This is expected, since blocks containing a larger number of transactions are usually larger in virtual size.

Another noteworthy result is the negative correlation of -0.40 between the number of transactions and the median fee rate. This suggests that blocks with higher transaction fees tend to contain fewer transactions.

The strongest observed correlation involving the Bitcoin price was with the number of transactions. However, since this analysis was performed on a relatively small dataset covering a period with limited Bitcoin price variation, this result is considered statistically insignificant.